

Human-Computer Interaction

Determining position using IMU (Inertial Measurement Unit).

The aim of the exercise is to explore the capabilities and operating principles of sensors (accelerometer, gyroscope, magnetometer) using a modern smartphone as an example.

Exercise 1 - Preparation of test scenarios

- Prepare several action scenarios.
- Try not to create too atomic or complex scenarios.
- A good scenario should have a name clearly indicating the actions being performed, e.g., "Scenario 1: Lifting the phone".
- Each scenario should include a description of the actions, e.g.:
 - The phone is lying freely.
 - Lift the phone smoothly to a height of 20 cm.
 - Return to the starting position at the same pace.
 - Lift the phone at the same pace to a height of 50 cm.
 - Return to the starting position at the same pace.
 - Lift the phone at the same pace to a height of 100 cm.
 - Return to the starting position at the same pace.
 - End.
- Sample scenarios should include: translations and rotations, individually and in combinations, at a constant and variable pace.

Exercise 2 - Set up an environment for acquiring data from a smartphone

- This task can be completed in various ways, ranging from recording sensor data directly on the smartphone to transmitting data in real time from the smartphone to a computer, where it can be visualized and analyzed in a programming environment (e.g., MATLAB, Python).
- The method of data acquisition will influence the final assessment of the task. Below are several example approaches to completing the task. Only one variant should be implemented. If someone decides to complete multiple variants, the most advanced one should be documented in the report.

- Variants for mobile data acquisition and visualization
 - Option 1 – Local acquisition and visualization on the smartphone
 - Purpose: Acquire sensor data and visualize it directly on the mobile device.
 - Tools: MATLAB Mobile, Phyphox, or any mobile application capable of recording and visualizing sensor data.
 - Instructions:
 - Install the chosen app on your smartphone.
 - Explore the basic functionality and configure the app to record and visualize sensor data (accelerometer, gyroscope, magnetometer) directly on the phone.
 - Perform a short test to ensure that data recording and visualization work correctly.
 - Option 2 – Local acquisition and further analysis on a computer
 - Purpose: Record data on the smartphone and perform offline analysis on a computer.
 - Tools: MATLAB Mobile, Phyphox, or any mobile application capable of recording sensor data to a file, and on the computer – any programming environment for further analysis (MATLAB, Python, etc.).
 - Instructions:
 - Record sensor data to a file on the smartphone.
 - Transfer the file to your computer.
 - Explore the file format and structure to understand how the data is organized.
 - Perform further processing and visualization using your preferred tool (MATLAB, Python, etc.).
 - Option 3 – Streaming data to a computer for real-time visualization
 - Purpose: Send sensor data from the smartphone to a computer and visualize it in real time.
 - Tools: Phyphox, MATLAB Mobile or any other.
 - Instructions for streaming data via HTTP using Phyphox:

- Configure the app to transmit data from selected sensors in real time. Note the IP address and port where the smartphone shares the data.
- Check in a web browser on your computer if you can access the data, e.g., `http://192.168.0.10:8080/`
- Acquire the data in a programming environment such as Python. Below are useful libraries and commands:
- The `requests` library, which allows HTTP requests such as GET
- Example URL string for retrieving accelerometer data:

```
url = 'http://192.168.0.10:8080/get?accX&accY&accZ&acc'
```
- Fetching data from the Phyphox server:

```
response = requests.get(url, timeout=5)
```
- Decoding the data as JSON:

```
data = response.json()
```
- Alternatively, real-time data streaming can be done using MATLAB Mobile. Log in with the same account on both the smartphone and MATLAB on the computer. On the computer, create a `mobiledev` object (`m = mobiledev`) to establish the connection. On the smartphone, select the sensors to record, set the sampling frequency, and choose “Stream to: MATLAB”. Start the streaming on the phone, and the data will be received live on the computer, where it can be accessed for example using `[a, t] = accellog(m)` or `m.accellog`, and then visualized. This allows continuous monitoring and visualization of sensor data while recording on the smartphone.

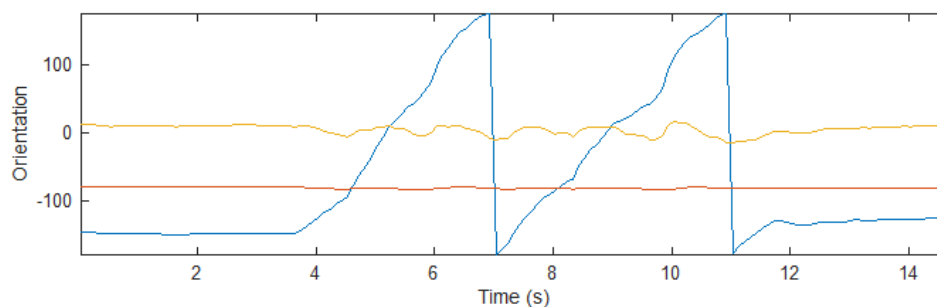
Exercise 3 - Perform the prepared scenarios

- Implement the developed scenarios by recording data from the sensors: accelerometer, gyroscope, and magnetometer.
- For each scenario, plot graphs showing the changes in the recorded parameters over time.
- Determine the range of changes in parameter values and explain what it means.
- Assess whether all events can be visually interpreted from the plots.
- Evaluate the effectiveness of each sensor in capturing the corresponding device movements.

- Analyze whether the accelerometer and gyroscope are equally effective in detecting translations and rotations.
- Pay attention to the sampling frequency during data acquisition. For a selected scenario, investigate how changing the sampling frequency affects the ability to detect events later — execute the scenario with several frequency variants.

Supplementary task - Optional

- Propose and implement a simple algorithm to count repetitive actions: number of steps, number of stairs climbed, number of rotations, number of squats...
- The algorithm should focus on calculating the number of local maxima for a selected sensor and axis. For example, consider data from the gyroscope in a scenario where the user performed two rotations around an axis while holding the smartphone in front of them.



Report

- Prepare a report on the exercises performed in PDF format.
- Send the report to: anowosielski@zut.edu.pl
- In the e-mail title, please use the scheme:
 - HCI Your Full Name